Evolutionary Optimization of Keyframe-Based Locomotion

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1 Introduction

The goal of this assignment is to develop an Evolutionary Algorithm (EA) to optimize keyframe-based locomotion for the **HalfCheetah-v5** robot in the MuJoCo physics environment via Gymnasium. Instead of a full policy, a cyclic sequence of fixed torque vectors (keyframes) drives the robot, and the EA evolves this sequence to maximize cumulative episode reward.

2 Representation

Each individual consists of $K = 5$ keyframes. Each keyframe is:

$$\mathbf{k}_i = [t_0,\ t_1,\ t_2,\ t_3,\ t_4,\ t_5,\ d]$$

where $t_j \in [-1.0,\ 1.0]$ are joint torques and $d \in [10,\ 50]$ is the duration in timesteps. Fitness is the total cumulative episode reward.

3 EA Configuration

Table 1: Hyperparameters

Parameter	Value
Keyframes (K)	5
Population size	20
Generations	30
Crossover	Single-point, $p_c = 0.8$
Mutation (torques)	Gaussian $\sigma = 0.15$, $p = 0.3$ per gene
Mutation (duration)	Integer ± 5, $p = 0.3$ per gene
Truncation pool	Top 10 (50%)
Seed	42

4 Selection Strategies

Three strategies taught in class were used in combination for parent and survivor selection.

Binary Tournament: Two random individuals compete; the fitter one is selected. Low-to-moderate selection pressure; preserves diversity well.

Truncation: Only the top 50% of individuals are eligible. Highest selection pressure; converges fast but risks premature convergence.

Fitness Proportionate Selection (FPS): Selection probability is proportional to fitness (shifted to handle negatives). Moderate pressure; maintains diversity across the population.

5 Experiments and Results

Four combinations of parent and survivor selection were tested. All experiments used single-point crossover, Gaussian mutation, population size 20, and 30 generations.

Table 2: Experiment Results

#	Parent Selection	Survivor Selection	Best Reward
1	Truncation	Truncation	116.78
2	Binary Tournament	Truncation	55.80
3	FPS	Truncation	**220.32**
4	Binary Tournament	FPS	-17.64

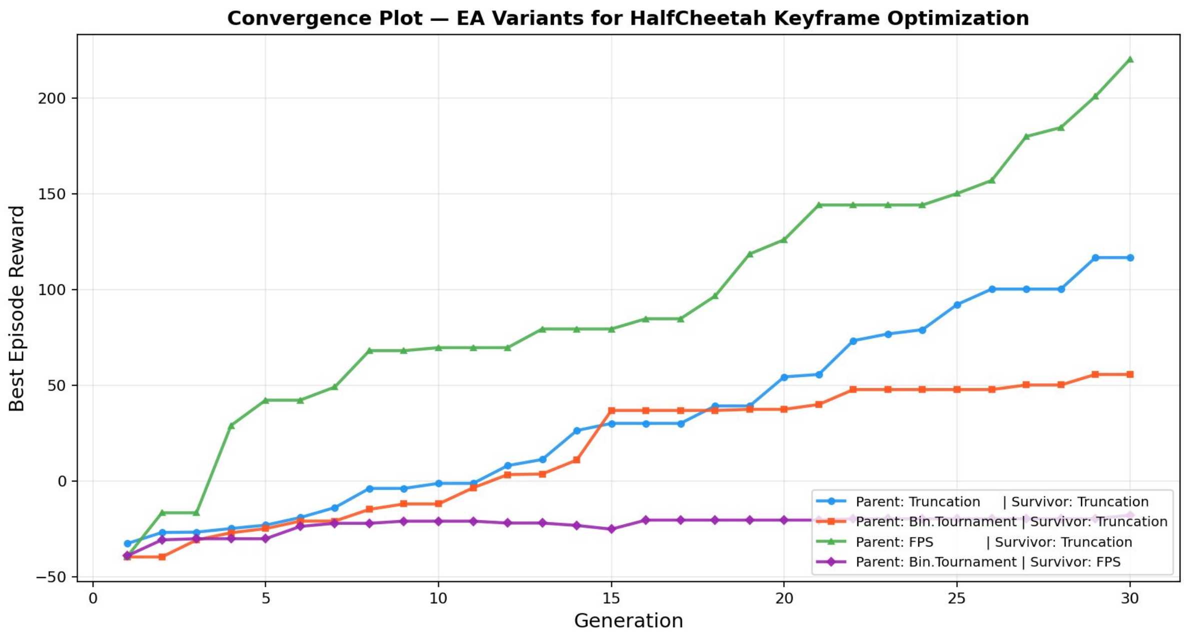

Figure 1: Best episode reward per generation for all four configurations.

Observations

- **Exp 1 — Truncation + Truncation (116.78):** Showed steady, consistent improvement across all 30 generations with no major jumps. The high selection pressure on both ends limited diversity but produced reliable convergence. Rewards crossed into positive territory around generation 12 and continued climbing steadily through to 116.78 by the end.
- **Exp 2 — Binary Tournament + Truncation (55.80):** Moderate performance overall. Binary tournament's lower selection pressure slowed early convergence, with rewards remaining negative until generation 11. The algorithm improved steadily in the second half but plateaued around 50 in the final generations, finishing at 55.80.
- **Exp 3 — FPS + Truncation (220.32):** The best performing configuration by a clear margin. FPS parent selection sampled broadly across the population proportional

to fitness, enabling strong and fast exploration. Combined with truncation survivor selection, the population quality improved rapidly — reaching positive rewards by generation 4 and crossing 100 by generation 19. The algorithm showed continued strong improvement all the way to generation 30, finishing at **220.32**. Visual evaluation confirmed strong locomotion with episode rewards of 215.70, 220.34, and 219.02.

- **Exp 4 — Binary Tournament + FPS (-17.64):** The worst performing configuration. FPS survivor selection preserves diversity but retains too many weak individuals when paired with binary tournament parents, which lack sufficient selection pressure to push the population toward better solutions. Rewards remained entirely negative throughout all 30 generations, never crossing zero, finishing at -17.64.

6 Best Keyframes Found

The best keyframes were produced by Experiment 3 (FPS + Truncation):

Listing 1: best_keyframes.txt

```
5
-0.7030, 0.7973, -0.3660, -0.5685, 0.3985, -0.1988, 10
0.1383, 0.0233, 0.0941, 0.1907, 0.3521, 0.1629, 27
0.2776, -0.2377, 0.1615, 0.2369, -0.2976, -0.4349, 10
-1.0000, -0.9888, -0.1711, 0.2570, 0.2395, 0.1347, 10
0.9085, -0.3813, -0.2339, 0.1104, 0.6053, 0.7700, 10
```

Total cycle length: $10 + 27 + 10 + 10 + 10 = 67$ timesteps. During visual evaluation this sequence achieved rewards of **215.70**, **220.34**, and **219.02** across three episodes.

7 How to Run

7.1 Prerequisites

Install the required packages once using:

Listing 2: Install dependencies

```
pip install gymnasium[mujoco] numpy matplotlib
```

7.2 Running the EA

Run `ea_halfcheetah.py` to execute all four experiments. The script prints generation-by-generation progress and saves outputs automatically.

Listing 3: Run the evolutionary algorithm

```
python ea_halfcheetah.py
```

After all experiments complete, two files are saved in the same folder:

- `best_keyframes.txt` — Best keyframe sequence found, in the required format (number of keyframes on line 1, then one keyframe per line with 6 torque values and duration).

- `convergence_plot.png` — Convergence chart showing best episode reward per generation for all four configurations. This also displays on screen automatically at the end of the run.

7.3 Visualising the Robot

A separate script `visualise_best.py` is provided to watch the optimized keyframes without re-running the full EA. It opens a MuJoCo render window showing the HalfCheetah robot moving with the best keyframes found.

Listing 4: Run the visualisation

```
python visualise_best.py
```

A MuJoCo window opens showing the 2D HalfCheetah robot on a chequered floor moving forward through the optimized keyframe cycle. The terminal prints the reward for each episode:

```
Episode 1: reward = 215.70
Episode 2: reward = 220.34
Episode 3: reward = 219.02
```

Note: Do not press any keys during the MuJoCo window as this triggers a screenshot save to `C:\tmp`. Ensure this folder exists beforehand by running `mkdir C:\tmp` in PowerShell, or simply close the window using the close button when done.

8 Conclusion

All four EA configurations started from negative rewards and three of the four successfully evolved positive-reward locomotion. The best configuration — FPS parent selection with Truncation survivor selection — achieved a final best reward of **220.32**, demonstrating that proportional parent sampling combined with strong survivor filtering is highly effective for this keyframe optimization task. The stark contrast between the best (220.32) and worst (-17.64) configurations highlights how critically the choice of selection strategy affects EA performance.